

Intro to AI Concepts

Data Science / AI Fundamentals

What Machine Learning and AI Actually Are

Artificial Intelligence is the broad field of building systems that perform tasks normally requiring human intelligence. Machine Learning is a specific approach to AI: instead of explicitly programming every rule, a system learns patterns directly from data. Deep Learning is a further subset of ML using neural networks with many layers, particularly effective on unstructured data like images, audio, and text. Understanding this nesting — AI contains ML, which contains Deep Learning as one approach within it — clears up a lot of confusion caused by these terms being used loosely and interchangeably in casual conversation and media coverage.

Common AI Application Areas

AI/ML techniques show up across a wide range of practical applications, each suited to different types of data and problems:

- Computer vision — image classification, object detection, facial recognition.
- Natural language processing (NLP) — text classification, translation, sentiment analysis, and the large language models behind modern chatbots.
- Recommendation systems — predicting what a user is likely to want next, based on their and similar users' past behavior.
- Predictive analytics — forecasting future values (sales, demand, risk) from historical data.

Responsible AI Basics

Deploying AI systems responsibly means being aware of several real risks: bias, where a model trained on historically biased data reproduces or amplifies that bias in its predictions (e.g. a hiring model trained on a company's past hires, if those hires skewed in some demographic direction); explainability, since some model types (especially deep learning) are effectively black boxes, making it hard to explain why a specific prediction was made — a real problem in high-stakes domains like lending or healthcare; and privacy, since training data often contains sensitive personal information that needs careful handling and, where required, anonymization or consent. Treating these as first-class engineering concerns, not afterthoughts, is table stakes for any AI system used on real people.

Where This Course Leaves You

This module intentionally stays introductory — a working conceptual map of the field rather than deep technical mastery of any one area. From here, the natural next steps for a student wanting to go further are: deeper statistics and linear algebra (the actual math underlying most ML algorithms), a specific application area (computer vision or NLP) studied in depth, and — most importantly — practicing on real, messy datasets rather than only clean tutorial data, since real-world data cleaning and problem framing is where most of an actual data scientist's time goes, not model training itself.

Learn More

Google's Machine Learning Crash Course

<https://developers.google.com/machine-learning/crash-course>

3Blue1Brown — Neural Networks Series

<https://www.youtube.com/@3blue1brown>

GeeksforGeeks — Machine Learning

<https://www.geeksforgeeks.org/machine-learning/>